Pediatric Respiratory Distress

AR-07 | Airway & Respiratory

Quick flow

01 Pediatric Distress [EMT / AEMT / Paramedic]

Position of comfort • appearance • work • air movement • SpO_2 • mental status • perfusion.

02 Wheeze, Stridor, or Another Cause? [EMT / AEMT / Paramedic]

Consider asthma, croup, bronchiolitis, epiglottitis, anaphylaxis, foreign body, or respiratory failure.

03 Initial Support [EMT / AEMT / Paramedic]

Least distressing effective oxygen method • suction when needed • pathway-specific bronchodilator/epinephrine.

04 Improving? [EMT / AEMT / Paramedic]

Reassess after each treatment; obtain access when severe or persistent.

05 Severe / Impending Failure? [AEMT / Paramedic]

PM-01 epinephrine when indicated and within provider scope • BVM support • request Paramedic airway escalation.

06 Paramedic - Severe Pediatric Asthma [Paramedic]

Medical-Control-directed corticosteroid; no standing pediatric steroid dose • magnesium when indicated • follow doses and safeguards below.

07 Reassess + Transport [EMT / AEMT / Paramedic]

Continuous SpO_2; $EtCO_2$ when indicated; maintain warmth and notify early.

Overview

- Assess pediatric respiratory distress by appearance, work of breathing, air movement, SpO_2, mental status, hydration/perfusion, and fatigue.
- Separate bronchospasm/asthma, croup/stridor, bronchiolitis, anaphylaxis, epiglottitis/bacterial tracheitis concern, foreign body, and respiratory failure early.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Allow position of comfort, minimize agitation, provide oxygen by the least distressing effective method, assist ventilation when tiring, and administer authorized albuterol.
- Treat anaphylaxis immediately through PM-01 and avoid airway manipulation when epiglottitis is suspected.
- For clear bronchiolitis, provide supportive care - suction, oxygen when needed, hydration assessment, and ventilation support for failure - rather than routine bronchodilator or epinephrine treatment.

AEMT

- Perform all EMT care plus vascular access, repeated albuterol and weight-appropriate initial ipratropium for moderate or severe bronchospasm, and PM-01-directed IM epinephrine. Request Paramedic care for nebulized epinephrine for croup/stridor; the nebulized route is Paramedic-only.

Paramedic

- Perform all prior care plus nebulized epinephrine for croup/stridor, Medical-Control-directed corticosteroid consideration, magnesium for severe asthma, cardiac/$EtCO_2$ monitoring, and BVM/advanced-airway support for impending failure.
- Move promptly to AR-05/AR-06 for silent chest, cyanosis, exhaustion, declining mental status, bradycardia, or ineffective ventilation; do not continue repeated medications in a tiring child.

Indications

- Pediatric patient with asthma, reactive airway disease, wheezing, bronchospasm, croup, stridor, bronchiolitis, or other respiratory distress.

Treatment steps

- Do not force a mask on an agitated child who is maintaining oxygenation. Use blow-by or the least distressing effective oxygen method; support ventilation immediately when oxygenation or ventilation is inadequate.
- For wheeze/bronchospasm, give albuterol and reassess after each treatment. Add ipratropium to the initial treatment only for moderate or severe bronchospasm.
- For croup/stridor, keep the child calm and avoid unnecessary airway manipulation. Paramedic-only nebulized epinephrine is indicated for stridor at rest or moderate/severe upper-airway obstruction. Transport after treatment because symptoms can recur.
- For suspected epiglottitis or bacterial tracheitis - drooling, tripod position, toxic appearance, or rapidly progressive upper-airway symptoms - do not force the child supine or examine the tongue/airway; prepare for a controlled airway response and notify the receiving facility early.
- For anaphylaxis or life-threatening asthma, use the approved PM-01 Pediatric Allergic Reaction epinephrine pathway; do not use a conflicting separate epinephrine dose in this protocol.
- For moderate or severe asthma/reactive-airway disease, contact Medical Control for corticosteroid consideration; it is adjunctive and must not delay bronchodilator therapy.
- Use magnesium sulfate for severe asthma not improving after initial bronchodilator therapy; monitor blood pressure.
- If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02.

Medications

Albuterol

Under 20 kg: 2.5 mg nebulized; 20 kg or greater: 5 mg nebulized for bronchospasm

Ipratropium

Under 20 kg: 0.25 mg nebulized; 20 kg or greater: 0.5 mg nebulized, with initial albuterol for moderate or severe bronchospasm

Nebulized epinephrine - Paramedic only

0.5 mL/kg of 1 mg/mL (1:1,000) epinephrine, maximum 5 mL (5 mg), nebulized for croup/stridor

Epinephrine 1 mg/mL (1:1,000)

Use the approved PM-01 Pediatric Allergic Reaction pathway for anaphylaxis or life-threatening asthma dosing

Methylprednisolone

Medical Control only; no standing pediatric dose, for moderate or severe asthma/reactive-airway disease

Magnesium sulfate

40-50 mg/kg IV once over 15-30 minutes; maximum 2 g, for severe asthma not improving after initial bronchodilators

Warnings / precautions

- Do not force a child supine; allow the position that best maintains the airway.
- Drooling, rapid onset, fever, and tripod positioning suggest epiglottitis or bacterial tracheitis; airway manipulation may cause complete obstruction.
- A silent chest is a pre-arrest finding.
- Do not routinely give albuterol or epinephrine for clear bronchiolitis.
- Do not delay airway support for repeated medications in a tiring child.
- Do not perform blind finger sweeps.

Clinical pearls

- A child with decreasing wheeze, worsening fatigue, or declining mental status may be deteriorating rather than improving.
- Stridor is an upper-airway finding; albuterol does not treat croup or fixed upper-airway obstruction.

Related pathways / contacts

PM-01 Pediatric Allergic Reaction - Open for anaphylaxis or life-threatening asthma requiring IM epinephrine.

AR-05 Pediatric Airway - Open for BVM and primary pediatric airway support.

AR-06 Pediatric Failed Airway - Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

PC-02 Pediatric Bradycardia / Poor Perfusion - Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.

AO-02 Newly Born - Open for neonatal respiratory support immediately after birth.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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